Moving averages is more frequently used for eliminating the seasonal variations. Even when applied for estimating trend values, the moving average method helps to establish a trend line by eliminating the cyclical, seasonal and random variations present in the time series. The period of the moving average depends upon the length of the time series data.

The choice of the length of a moving average is an important decision in using this method. For a moving average, appropriate length plays a significant role in smoothening the variations. In general, if the number of years for the moving average is more then the curve becomes smooth.

Merits

- It can be easily applied
- It is useful in case of series with periodic fluctuations.
- It does not show different results when used by different persons
- It can be used to find the figures on either extremes; that is, for the past and future years.

Demerits

- In non-periodic data this method is less effective.
- Selection of proper 'period' or 'time interval' for computing moving average is difficult.
- Values for the first few years and as well as for the last few years cannot be found.

Moving averages odd number of years (3 years)

To find the trend values by the method of three yearly moving averages, the following steps have to be considered.

1. Add up the values of the first 3 years and place the yearly sum against the median year. [This sum is called moving total]
2. Leave the first year value, add up the values of the next three years and place it against its median year.
3. This process must be continued till all the values of the data are taken for calculation.
4. Each 3-yearly moving total must be divided by 3 to get the 3-year moving averages, which is our required trend values.

Example 7.5

Calculate the 3-year moving averages for the loans issued by co-operative banks for non-farm sector/small scale industries based on the values given below.

Year	2004-05	2005-06	2006-07	2007-08	2008-09	2009-10	2010-11	2011-12	2012-13	2013-14	2014-15
Loan by District Central Cooperative banks (Rupees in crores)	41.82	40.05	39.12	24.72	26.69	59.66	23.65	28.36	33.31	31.60	36.48

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